Pranshu Nijhawan

Principal Architect - Agentic AI & Platform Architecture

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Summary

Principal Architect with 10+ years in enterprise SaaS. Sole architecture owner of **Promotion Advisor**, a flagship **multi-tenant SaaS** in McKinsey's **Periscope** suite, covering its planning application, Analytics Engine, and agentic capabilities. Recent work spans agentic architecture across an **MCP mesh** (**tool discovery and routing**, deterministic business logic, **agent data-access control**) and high-performance data systems (sub-2 second response on 2M+ records, billion-row analytics on **ClickHouse**, **semantic vectorized lookup** over **JSONB**). Coached org-wide **Cursor** and AI-assisted engineering practices for a 37% capacity increase, and led the shift to **Spec-Driven Development** and an **AI-led SDLC**.

- **Platform AI**: co-built **Periscope**'s enterprise agent platform, the Git-versioned agent definition, harness, and gateway layer that every product agent now deploys through.
- **Cortex**: a stateless engine that compiles configurable business rules into deterministic executable binaries, so agent decisions are version-controlled and auditable instead of authored in prompts.
- **Context-Scoped Catalog**: an intelligent **selection algorithm** that keeps agent context bounded and **tool discovery** accurate to the context however large the enterprise tool fleet grows.

Experience

McKinsey & Company *March 2022 - Present*

Principal Architect I

Previously: Senior Software Engineer II, Senior Software Engineer I

> *Sole architecture owner of Promotion Advisor, a flagship* **multi-tenant SaaS** *in McKinsey's Periscope suite, covering the planning application, the Analytics Engine, and the platform's agentic capabilities. Co-built* **Platform AI**, **Periscope***'s enterprise agent platform, and designed the agentic layer that runs on it:* **Cortex** *for deterministic business logic and a* **context-scoped catalog** *for tool selection at fleet scale. Authored the platform's first unified architecture reference and set the cost and API standards now replicated by sibling* **Periscope** *products.*

- **Platform Architecture Ownership:** Own the architecture of **Promotion Advisor** end to end: the planning application (**React-**, **TypeScript**, **Node.js**, **GraphQL**, **C# / .NET microservices**), the Analytics Engine (**Databricks**, **PySpark**, **DBT**, **PostgreSQL**), and the agentic layer. Accountable for the configuration framework (**GitOps**-based, zero-downtime configuration delivery), the security model, polyglot persistence (**PostgreSQL**, **Redis**, **Databricks Delta tables**, **ClickHouse**), ETL, the **Periscope Reporting Engine**, and the external API surface. Set technical direction, ADRs, and release architecture across five delivery squads.
- **Platform AI: Enterprise Agent Platform:** Co-built **Platform AI**, **Periscope**'s enterprise agent platform: a **GitOps Agent Definition Manager** that versions every agent as Git artifacts, not prompt text; a **LangChain** harness on Kubernetes running one agent loop for all product agents; and a **Tenant MCP** gateway owning authentication, **tool discovery**, and quota. Caller identity survives every hop.
- **Cortex: Deterministic Business-Logic Engine:** Built **Cortex**, a stateless engine that runs business logic written as configuration. Rules compose from three operations (**acquire**, **derive**, **assert**) into **playbooks**; a purpose-built **TypeScript** compiler parses the typed YAML into executable binaries. Business rules stay deterministic and unit-testable, so agent output is reproducible and auditable.
- **Context-Scoped Catalog: Tool Selection at Fleet Scale:** Solved **context pollution**: an agent cannot reach a large **tool estate** if every tool is bound up front. Designed the **selection algorithm** (pinned tools, a free **BM25** shortlist, then a cheap router model for the near-duplicates) and scoped the catalog per tenant, which makes version gating exact and routing deterministic. Version-mismatch failures dropped to zero, and a new solution registers without per-tool wiring.
- **Periscope Reporting Engine on ClickHouse & Real-Time Conflict Detection:** Built the **Periscope Reporting Engine** on **ClickHouse** (**Projections**, **Dictionaries**, billion-row sub-second queries): users compose their own reports and every aggregation resolves at request time against a governed **ontology** through a dynamic **Query Builder**, which removes the batch-prep step and the analytics-team dependency for each new measure. Also designed the multi-dimensional promotion conflict detection algorithm, progressive candidate narrowing over sorted item arrays: sub-2s at 20 concurrent users across 2M promotions, 5,000x faster on large item groups.
- **AI-Assisted Engineering & Agent Tooling:** Drove org-wide adoption of **Cursor** and **Spec-Driven Development** (**OpenSpec-**, **SpecKit**), which raised team capacity 37%. Built the agent library behind it: 10+ production agents for architecture diagrams, ADR and APR authorship, and technical communications, plus an SDD Agent that runs an 11-dimension anti-hallucination audit per spec. The **Repo Mastery Agent** replaced a heavy **Graph RAG** documentation system with a **Registry Pattern** over vector binaries and explicit dependency relations.

Technologies: MCP, LangChain, Cursor, OpenSpec, SpecKit, BMad, PyTorch, T5, TypeScript, Python, Node.js, Fastify, C#, .NET, React, GraphQL, PostgreSQL, ClickHouse, Databricks, Casbin, Kubernetes, Azure, AWS

Eptura (formerly Condeco) *April 2021 - March 2022*

Principal Engineer

> *Re-architected a single-tenant workspace SaaS to cloud-native microservices on AKS and built a global IoT release orchestrator that pushes firmware to 300M+ devices. Directed 20 engineers across four squads.*

- **Cloud-Native Re-Architecture:** Migrated legacy single-tenant SaaS to cloud-native microservices on AKS (Azure Kubernetes Service): 60% faster incident recovery, eliminated single-tenant scaling bottlenecks.
- **Global IoT Release Orchestration:** Built a release orchestration system (Azure IoT Hub, Azure Functions, Azure CosmosDB) that delivers firmware updates to 300M+ IoT devices worldwide with zero-downtime rollouts.
- **Engineering Leadership:** Directed 20 engineers (4 tech leads) across four squads. Established a structured mentorship program: +20% retention, accelerated engineering progression.

Technologies: C#, .NET, React, Angular, AKS, Azure IoT Hub, Azure Functions, Azure CosmosDB, Azure, SQL Server

Nagarro *December 2019 - April 2021*

Senior Engineer

> *Shipped full-stack web and mobile apps for e-commerce and HR enterprise clients on Angular, .NET Core,* **PostgreSQL***, and Ionic. CI/CD and IaC practices cut deployment failures by 70% across client projects.*

- **Full-Stack Web + Mobile Delivery:** Delivered production web and mobile apps on Angular, .NET Core, **PostgreSQL**, and Ionic Framework at 99.5%+ uptime.
- **CI/CD & Infrastructure as Code:** Established CI/CD pipelines and IaC practices: 70% reduction in deployment failures across client projects.

Technologies: C#, .NET Core, Angular, Ionic Framework, PostgreSQL, DynamoDB, AWS, Fargate

Eptura (formerly Condeco) *September 2018 - December 2019*

Software Engineer

> *Built* **C#** */ .NET /* **React** */ SQL Server microservices that carried 3x customer growth without reliability regressions. Cut build times 40% and core query latency 3x through data model refactoring.*

- **Microservices for 3x Customer Scale:** Designed and shipped **C#** / .NET / **React** / SQL Server microservices that carried 3x customer growth without reliability regressions.
- **CI/CD & Data Model Optimization:** Cut build times by 40%; refactored core data models to reduce ave**rag**e query latency by 3x.

Technologies: C#, .NET, React, Angular, SQL Server, Azure

Gartner *February 2016 - August 2018*

Associate Software Engineer

Previously: Intern

> *Built backend features on* **C#** */ .NET Core /* **React** *for an HR analytics SaaS, which improved survey data accuracy 60% for consulting clients. Automated SQL workflows saved 240+ engineering hours annually.*

- **SaaS Analytics Backend:** Shipped backend features on **C#** / .NET Core / **React**: improved survey data accuracy by 60% for HR consulting clients.
- **SQL Workflow Automation:** Automated SQL workflows (stored procedures, SQL Agent jobs): saved 240+ engineering hours annually.

Technologies: C#, .NET Core, React, SQL Server, AWS

Core Competencies

– **AI & Agentic Engineering:** Agentic AI Architecture, Multi-Agent Orchestration, MCP (servers and clients), Tool Discovery and Routing, Context Engineering, Deterministic Business-Logic Compilation, Agent Data-Access Control, RAG (Retrieval-Augmented Generation), Cursor (Rules, Skills, Agents, Hooks, SDK), Spec-Driven Development (OpenSpec, SpecKit, BMad).

– **SaaS & Architecture:** System Design, Microservices, Distributed Systems, Domain-Driven Design (DDD), API Design (REST, GraphQL), Configuration-Driven Architecture, Polyglot Persistence, Multi-Tenant SaaS, ADR / APR Authorship.

– **Cloud & Infrastructure:** Azure, AWS, Kubernetes (AKS), Terraform, GitOps, CI/CD, Zero-Downtime Delivery, Observability (Dynatrace, OpenTelemetry).

– **Data Engineering & Real-Time Analytics:** Databricks, Spark (PySpark), DBT, ClickHouse, PostgreSQL, Redis, Real-Time Analytics, ELT / ETL, Columnar Analytics, FinOps / Cost Optimization.

– **Tech Stack:** TypeScript, Python, C#, .NET Core, Node.js, Fastify, React, Angular, SQL.

Certifications

- Microsoft Certified: Azure Data Engineer Associate
- Microsoft Certified: Azure Developer Associate

Educational Qualifications

Gurgaon Institute of Technology & Management **2012 - 2016**
Bachelor of Technology (B. Tech), Computer Science & Engineering